

Revisiting Perfusion Assays under Wearable Constraints

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Abstract

We examine the behaviour of electrode arrays in resource-limited clinics. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. The measurements reported here were collected over twelve independent trials under controlled conditions.

1. Introduction

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Taken together, these findings argue for treating the two regimes separately in future analyses. Sensitivity to the smoothing window proved modest across the range we examined. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The reduced model retains most of the explanatory power at a fraction of the computational cost. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

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2. Related Work

Neither of the alternative formulations improved on the simple additive model in our setting. Residual error grows approximately linearly with the number of coupled parameters. Replicate agreement was high, suggesting the protocol is robust to small handling differences. The reduced model retains most of the explanatory power at a fraction of the computational cost. Taken together, these findings argue for treating the two regimes separately in future analyses. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Longer observation windows narrow the confidence interval but do not shift the central estimate. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

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3. Materials and Methods

Sensitivity to the smoothing window proved modest across the range we examined. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Calibration drift over the trial period remained below the threshold set out in the protocol. Neither of the alternative formulations improved on the simple additive model in our setting. We release the processing scripts and intermediate artefacts to support independent verification. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Residual error grows approximately linearly with the number of coupled parameters.

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4. Experimental Setup

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